Applied Algebra

Spring 2026

Instructor: Tuan Tran

Exercise Sheet 8

Deadline: by the end of next Wednesday

Instructions. You are encouraged to work in pairs. When submitting your solutions, list the team members and circle the name of the person responsible for writing up the solutions that week.

Please submit written solutions to **TWO** of the following problems. **Justify your answers.**

Problem 1. Which of the following are rings? For those which are, decide whether they have identity elements, whether they are commutative, and whether they have zero-divisors.

(i) The set of all 3×3 upper-triangular matrices with real entries, under the usual matrix addition and multiplication.

(ii) The set of all upper-triangular real matrices with 1's along the diagonal, of the form

$$\begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix},$$

with the usual operations.

(iii) The set $\mathcal{P}(X)$ of all subsets of a set X, with intersection as multiplication and union as addition.

Problem 2. Use the definition of a ring to prove the following. For all x, y, z in a ring,

(i) $x(-y) = -xy = (-x)y$,

(ii) $(-1) \cdot x = -x$,

(iii) $(x - y)z = xz - yz$ (where $x - y$ means, as usual, $x + (-y)$).

Problem 3. Which of the following are vector spaces over the given field?

(i) The set of all 2×2 matrices over $\mathbb{R}$, with the usual addition, and scalar multiplication given by

$$\lambda \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} \lambda a & \lambda b \\ \lambda c & \lambda d \end{pmatrix}.$$

(ii) The set of all 2×2 matrices over $\mathbb{R}$, with the usual addition, and scalar multiplication given by

$$\lambda \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} \lambda c & a \\ b & \lambda d \end{pmatrix}.$$

(iii) The set of all 2×2 matrices over $\mathbb{R}$, with the usual addition, and scalar multiplication given by

$$\lambda\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} \lambda^{-1}a & b \\ c & \lambda^{-1}d \end{pmatrix} \qquad \text{for } \lambda \neq 0,$$

and

$$0\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Problem 4. Let F be any field. The characteristic of F is defined to be the least positive integer n such that

$$\underbrace{1+1+\cdots+1}_{n \text{ times}} = 0.$$

If there is no such n, then the characteristic of F is said to be 0. Show that if the characteristic of F is not zero, then it must be a prime number.